

Screen Capture Rectification and Temporal Stabilization for Real-world Captured-screen Videos

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Problem. Existing screen demoiré datasets validate the demand for captured-screen restoration, but their inputs are often controlled, cropped, paired, or spatially/temporally aligned. Real phone-captured screen videos still contain background regions, perspective distortion, hand-held camera shake, weak screen borders, and dynamic screen content. This project targets this missing pre-alignment stage.

Task and Goal

Input is a full camera video of a computer screen. Output is a fixed-ratio, front-facing, temporally stable screen video. The goal is to remove most non-screen background, rectify perspective distortion, reduce frame-to-frame jitter, and provide stable screen-coordinate input for downstream video demoiré, OCR, or archival restoration.

Method

- Detect or manually specify the first-frame screen corners.
- Estimate a homography to rectify the screen to a 16:9 canvas.
- Track reference-plane features with Lucas-Kanade optical flow.
- Use RANSAC and geometric gates to reject unreliable updates.
- Interpolate/smooth the trajectory and evaluate residual motion.

Dataset Plan

The final evaluation will use a self-collected test set: **5 scenario classes x 10 clips x about 5 s**. The classes are static pages/documents, scrolling webpages, in-screen video playback, PPT or weak-border low-texture pages, and 4K/moiré/glare hard cases. Public demoiré datasets are used as related-work evidence rather than direct benchmarks for this pre-alignment task.

Metrics

We estimate residual affine motion between adjacent normalized frames. Main metrics are residual translation p95, residual rotation p95, and residual scale-delta p95. Tracker accept ratio, RANSAC inliers, inlier ratio, and feature coverage are used to explain failure cases. Lower residual motion means a more stable normalized video.

Initial Experiment

One local 1920x1080 captured-screen video with 317 frames was used for an initial same-video ablation. The numbers below measure residual motion in the normalized output; they are not ground-truth reconstruction errors.

Method	Last 2 s translation p95	Last 2 s rotation p95	Interpretation
Frame-wise corner detection	1.927 px	0.0425 deg	jittery baseline
Optical-flow tracking	1.929 px	0.0263 deg	lower rotation, still unstable
Reference-plane tracking	0.118 px	0.0044 deg	best residual stability

Expected Result and Timeline

The expected result is a classical geometric preprocessing pipeline that converts realistic captured-screen videos into stable, rectified screen-coordinate videos. Proposal-stage work has fixed the application story, implemented the core normalization pipeline, and produced an initial ablation. Final-stage work will complete the 50-clip test set, run method ablations, visualize corners and tracking failures, and report both successful and unstable cases.

Source files and evidence are stored under deliverables/proposal_20260622/. Current draft uses placeholder Group ID until the official group number is filled.